Recap Checkpoint 5 — Cumulative: Component 1 (1.1, 1.3, 1.4) + Component 2 (2.1–2.3, incl. Algorithms)

Name: ____ Date: _ ***Mark:*** ___ / 35

OCR H446 cumulative recap — mixes every topic taught so far. Show all working.

Questions

Q1 [3] (AO1). *(1.1 — Architecture)* Compare **Von Neumann** and **Harvard** architectures by stating **one** difference in how they store instructions and data, and giving **one** advantage of each.

Q2 [4] (AO2). *(1.4 — Calculations)* Show all working.

(a) Convert hexadecimal **3B** to denary. **[2]**

(b) A bitmap image is 64 × 48 pixels with a colour depth of **4 bits per pixel**. Calculate its size in **kibibytes (KiB)**, ignoring metadata. **[2]**

Q3 [4] (AO2). *(1.4 — Boolean algebra)*

(a) Simplify the Boolean expression **A·B + A·B̄** and name the law(s) used. **[2]**

(b) Complete the output column of the truth table for **Q = (A OR B) AND NOT B**. **[2]**

A	B	Q
0	0	
0	1	
1	0	
1	1	

Q4 [3] (AO1). *(1.3 — Databases)* A relational database has tables `Student` and `Course`.

(a) State what is meant by a **foreign key**. **[1]**

(b) Explain how **normalisation to 3NF** reduces data redundancy. **[2]**

Q5 [4] (AO2/AO1). *(2.2 — Programming paradigms)*

(a) State **one** key difference between **procedural** and **object-oriented** programming. **[1]**

(b) Define **encapsulation** and explain **one** benefit it provides. **[2]**

(c) State what is meant by **inheritance**. **[1]**

Q6 [4] (AO2). *(2.3 — Searching/sorting trace)* A list contains `[8, 3, 5, 1]`. Carry out a **merge sort**, showing the lists produced at **each** split and merge stage. State the **worst-case** time complexity of merge sort.

Q7 [4] (AO1/AO2). *(2.3 — Big O)*

(a) State the worst-case time complexity (Big O) of: linear search, binary search, and bubble sort. **[3]**

(b) An algorithm contains a single loop that runs `n` times, and inside it a binary search over the same `n` items. State its overall time complexity. **[1]**

Q8 [4] (AO2). *(2.3 — Graphs/trees)* For the binary search tree below, give the **in-order** traversal and state what property it demonstrates, then state the **order** in which `27` would be compared against existing nodes if inserted.

```
        40
       /   \
     20     55
    /  \      \
  10   30     70
```

Q9 [5] (AO3). *(Synoptic extended response — 2.1/2.2/2.3)* A music-streaming service must let users search a library of millions of songs by title and play results quickly.

Explain why the library should be **kept sorted** and a **binary search** used rather than a linear search, referencing **time complexity**, and identify **one** trade-off introduced by keeping the data sorted. **[5]**